

AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

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TL;DR

The final layer of a vision foundation model is **not** always the best representation. AMDF learns, for **every patch**, how to mix features from **all frozen layers**, with no layer search.

- **AMDF = depth-aware fusion**: 16 learned expert queries attend over all L layers per patch; backbone frozen, no bottleneck.
- **Best mean rank** on surface normals (all 7 backbones), segmentation and Taskonomy; second on depth and camera pose, behind ConcatLayers, at 2–4× fewer trainable parameters (1.6M vs 6.8M on DINOv3-L).
- Its learned depth-attention is interpretable, and recovers the same semantic–geometric split across depth found via exhaustive oracle layer sweeps.

Motivation & Overview

Representations change abstraction level across network depth: low-level geometry, mid-level scene structure, and high-level semantics peak at different layers. AMDF removes the need to hand-pick a layer — it reads all frozen layers and, for **every patch**, learns how to mix them with depth attention.

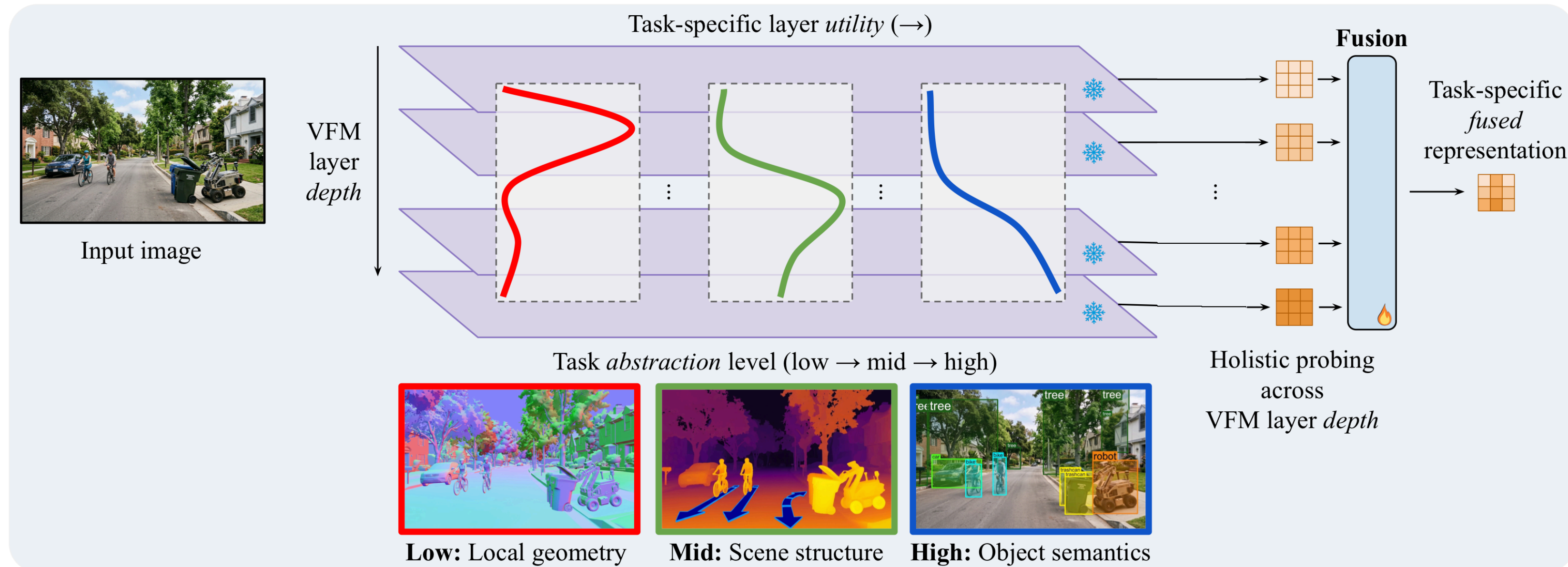


Fig. 1: Task-specific layer utility varies with abstraction level (low/mid/high). AMDF takes the frozen patch tokens of every layer and fuses them per patch, with learned depth attention, into a task-specific dense representation.

Abstraction–Depth Correlation: Empirical Evidence

We probe a frozen DINOv3-L/16 independently at every layer $\ell \in \{1, \dots, 24\}$ on three NYUv2 tasks of increasing abstraction, training a lightweight head per layer and recording its validation score. All three tasks share the same training images, yet the best-performing layer shifts systematically deeper as task abstraction increases.

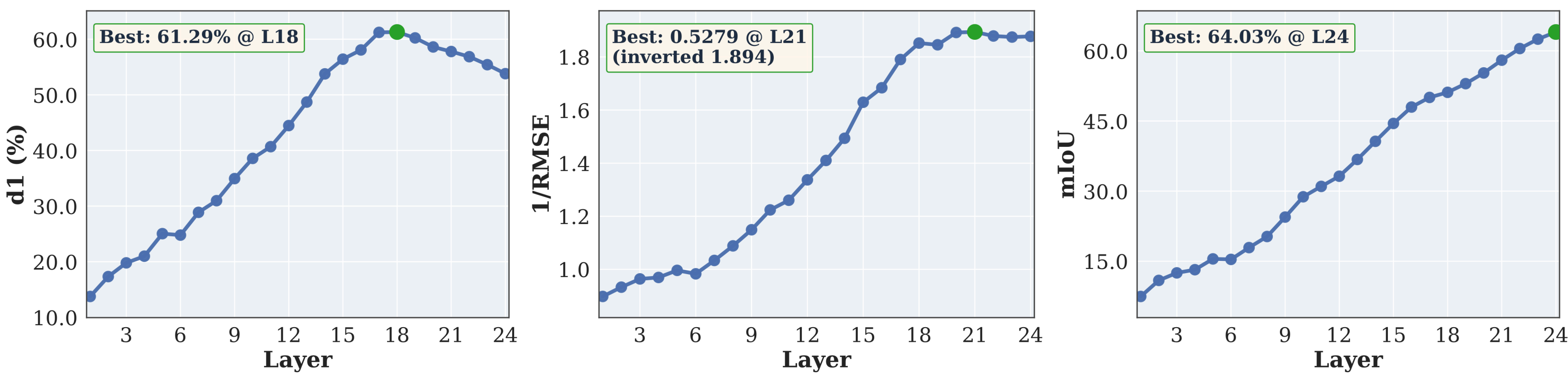


Fig. 2: Layer-wise probing on NYUv2 with DINOv3-L/16: the best layer shifts deeper as tasks become more semantic — surface normals peak at layer 18, depth at layer 21, segmentation at the final layer 24 (the last layer).

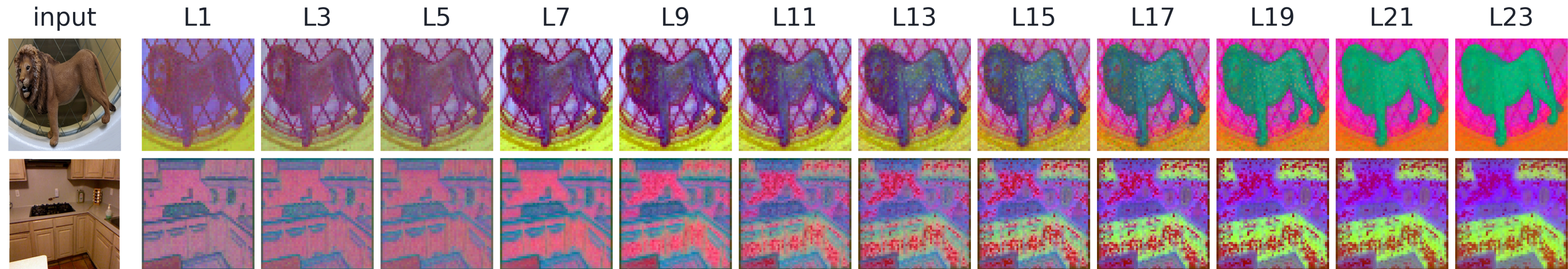


Fig. 3: Frozen PESpatial-L patch features (one PCA per image, 770 px input): object-background separation emerges only in deep layers ($\geq$ L15).

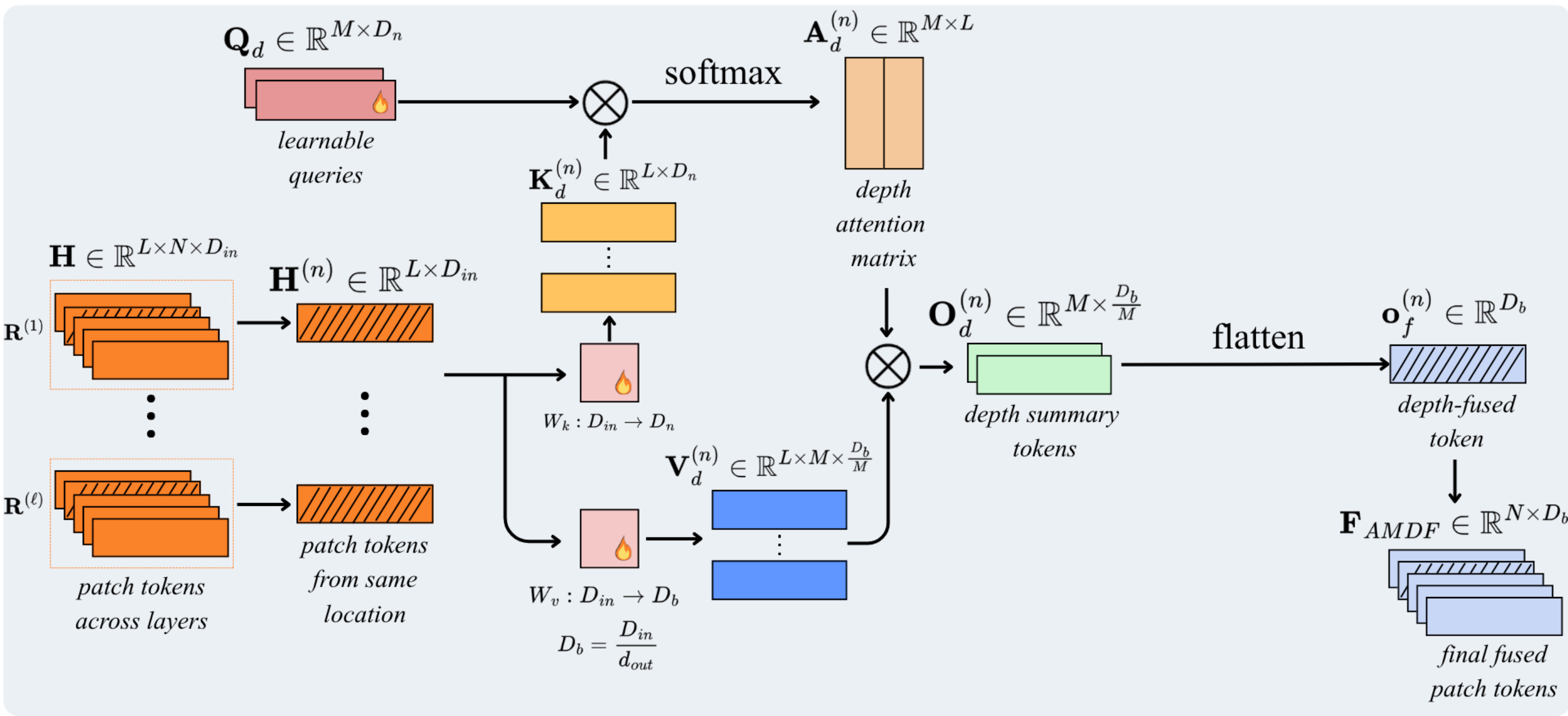
Exhaustive per-task layer search is costly and still commits to one *fixed* depth. Worse, a single layer ignores complementary information spread across *multiple* depths, and concatenating all layers is strong but expensive (6.8M vs 1.6M trainable parameters on DINOv3-L, including the task head). This motivates AMDF's patch-wise fusion across the full frozen hierarchy.

Conclusion

- Useful features are **spread across depth**, and the best depth depends on the task.
- AMDF has a better mean rank than LastLayer, AvgLayer and WeightedLayers in every table. It leads ConcatLayers on normals, segmentation and Taskonomy, and comes close on depth and pose with far fewer parameters.
- On fine-grained classification, a dedicated last-layer probe (EfficientProbe) remains stronger.
- Study VFMs as **full hierarchies**; no manual layer search needed.

Method: Adaptive Multi-Layer Dense Fusion

Depth-aware fusion — for every patch, each layer's token is projected to a key (with a learned layer embedding) and to a value of D/2 channels split into 16 expert blocks. Each of 16 learned expert queries takes a softmax over all L layers and mixes its block of the values, so every expert picks its own depth per patch.



Implementation & Task Heads

- The VFM backbone stays **frozen** throughout; only the lightweight fusion probe and the downstream task head are trained end-to-end.
- All available ViT blocks are exposed to AMDF: no manual subset selection or hand-picked layer depth is required.

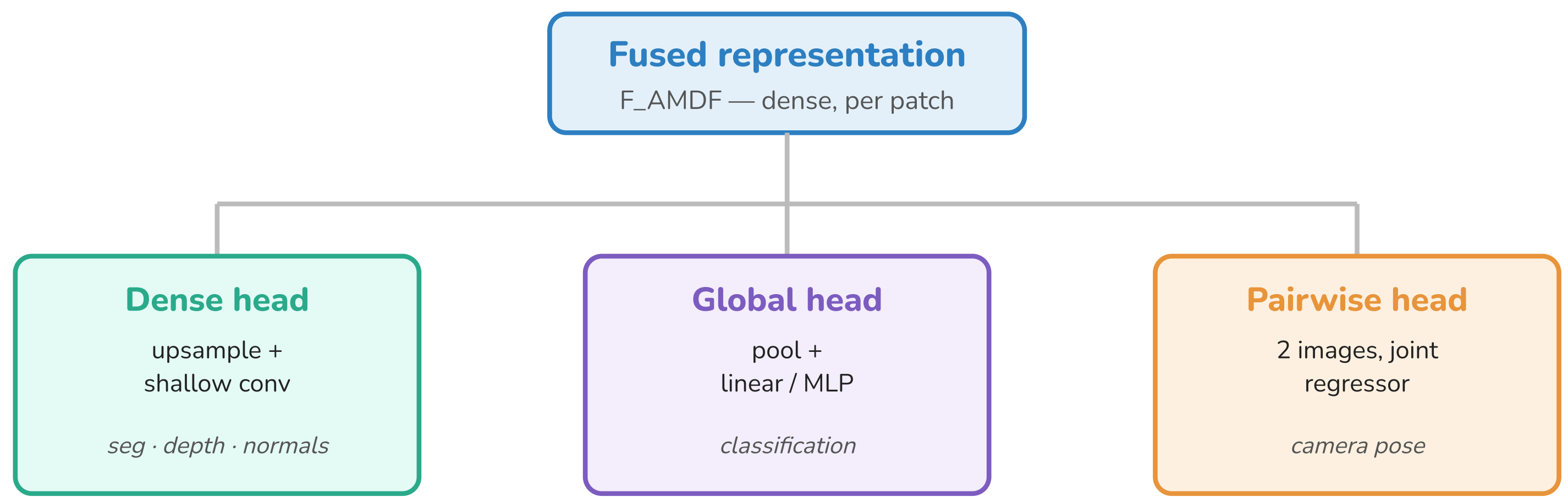


Fig. 4: The same frozen-backbone fusion module feeds three task-specific heads — only the head (and its supervision) changes per task.

Learned depth-attention tracks layer utility

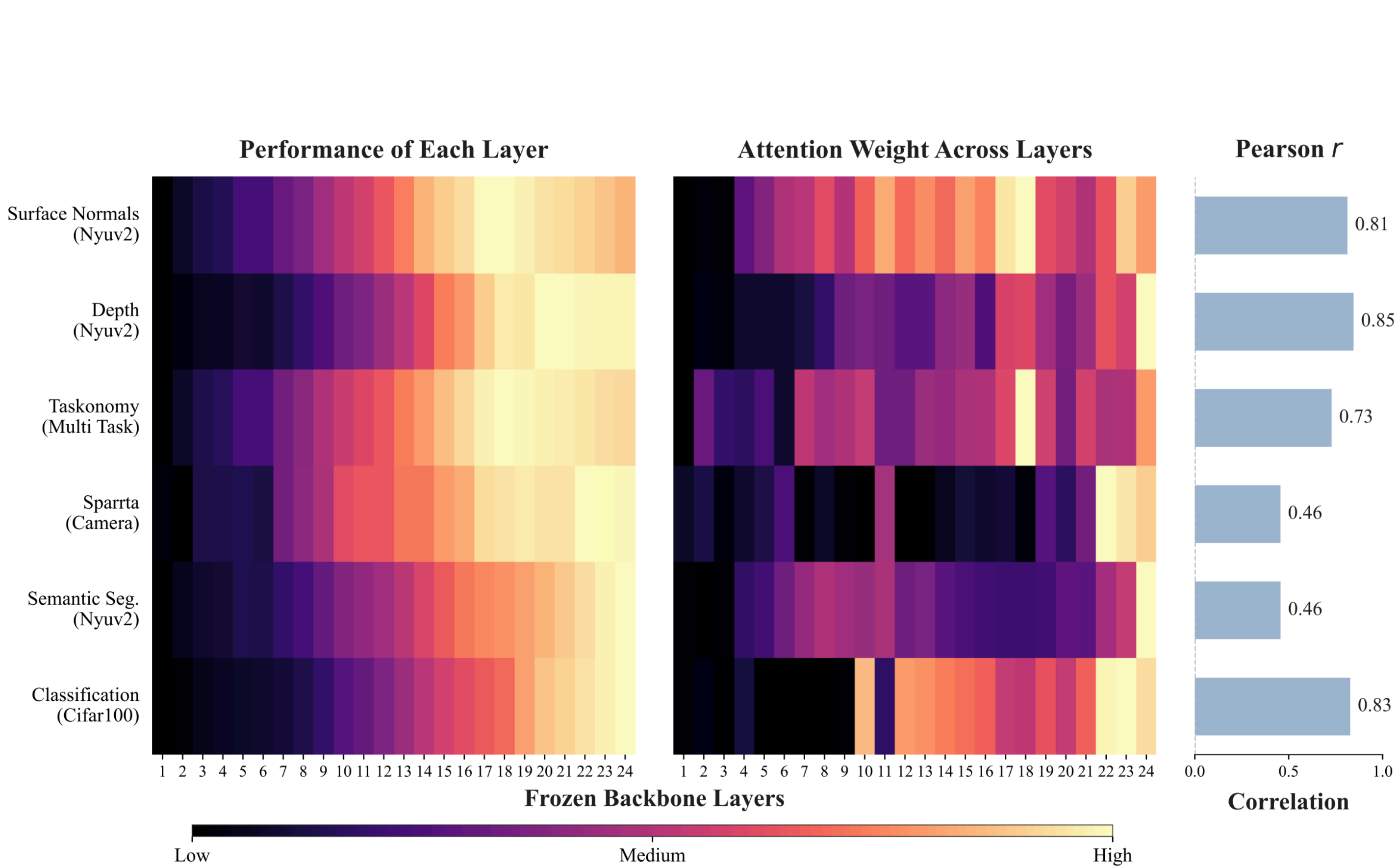


Fig. 5: AMDF's learned depth-attention (middle) tracks per-layer probe performance (left) on DINOv3-L/16. Alignment (Pearson r , right) is strongest for dense geometric tasks (depth $r=0.85$, normals $r=0.81$, Taskonomy $r=0.73$) and shifts toward deep layers for semantic tasks (classification $r=0.83$).

Results on NYUv2 (increasing abstraction)

Params (M): trainable parameters (probe + task head) with a ViT-L backbone (D=1024, 24 layers — DINOv3-L, MAE-L, SigLIP2-L, V-JEPA2.1-L, PESpatial-L).

Surface normals — d_1 ($\uparrow$)

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	22.22	24.55	23.66	32.84	13.23	36.31	16.88	0.53	4.57
AvgLayer	20.72	27.84	31.01	35.34	15.24	32.55	17.35	0.53	4.29
WeightedLayers	25.45	30.68	34.61	36.54	16.81	35.50	17.48	0.53	3.14
ConcatLayers	30.91	35.21	39.26	41.00	24.71	39.57	22.46	6.82	2.00
AMDF (ours)	35.91	40.75	45.19	46.96	29.01	45.48	28.38	1.62	1.00

Depth estimation — d_1 ($\uparrow$)

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	60.79	65.27	66.44	62.49	58.40	69.14	56.90	0.66	4.00
AvgLayer	53.39	63.70	65.50	65.82	61.21	66.92	58.95	0.66	4.57
WeightedLayers	57.60	65.21	66.66	66.45	61.27	68.53	59.03	0.66	3.43
ConcatLayers	65.10	70.72	73.21	69.28	66.36	71.73	61.42	6.95	1.14
AMDF (ours)	62.42	69.30	71.83	68.23	63.84	71.76	59.46	1.75	1.86

Semantic segmentation — mIoU ($\uparrow$)

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	43.89	53.38	57.19	39.08	40.85	44.11	32.88	0.54	3.21
AvgLayer	33.70	47.37	50.89	40.69	42.18	42.10	33.42	0.54	4.29
WeightedLayers	39.30	50.17	53.07	41.30	42.13	43.11	33.40	0.54	3.86
ConcatLayers	42.92	51.46	54.88	41.44	43.36	43.98	34.12	6.84	2.43
AMDF (ours)	44.04	53.38	57.20	42.05	43.39	45.26	33.49	1.64	1.21

Shared multi-task representation (Taskonomy)

Aggregate $L_1 \times 100$ ($\downarrow$)

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	41.64	39.21	39.27	39.52	46.81	36.97	42.23	0.53	4.43
AvgLayer	43.66	39.42	38.50	38.55	45.51	37.84	41.82	0.53	4.43
WeightedLayers	41.46	38.48	37.49	38.14	44.40	37.27	41.68	0.53	3.14
ConcatLayers	37.76	35.19	33.62	34.54	38.74	33.45	37.64	6.82	1.71
AMDF (ours)	38.82	34.64	32.69	34.13	37.90	35.10	37.58	1.62	1.29

Fine-Grained Classification (CUB-200, Top-1)

CUB-200 — fine-grained bird recognition, Top-1 ($\uparrow$)

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
AvgLayer	41.19	60.29	54.71	37.41	81.45	29.96	41.39	0.63	6.29
WeightedLayers	64.19	77.83	73.01	45.30	81.11	33.60	43.43	0.63	4.71
ConcatLayers	56.92	73.40	71.67	49.83	81.71	41.38	42.72	6.92	4.57
AvgPool	62.70	78.19	71.79	25.58	79.66	27.05	39.81	0.20	5.86
EfficientProbe	77.22	84.01	86.19	74.38	83.95	74.81	71.94	1.27	1.14
AttentiveFusionAP	64.53	77.98	77.51	50.17	81.99	35.80	44.17	8.60	3.00
AMDF (ours)	65.25	78.49	75.23	47.13	84.62	35.88	59.99	1.72	2.43

NAVI Camera Pose (Pairwise Head)

Method	DINOv3-S		DINOv3-B		DINOv3-L		MAE-L		SigLIP2-L		V-JEPA2.1-L		PESpatial-L		Params (M)	Rank
	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.		
LastLayer	45.49	0.17	44.58	0.17	40.33	0.16	38.74	0.16	62.69	0.22	37.05	0.15	65.22	0.23	1.22	4.25
AvgLayer	43.97	0.17	36.13	0.15	35.44	0.14	36.73	0.15	59.33	0.21	47.21	0.18	66.16	0.23	1.22	3.96
WeightedLayers	43.31	0.17	35.28	0.15	35.06	0.14	34.88	0.14	58.86	0.21	43.40	0.17	67.01	0.24	1.22	3.43
ConcatLayers	36.71	0.15	30.30	0.13	28.37	0.12	31.74	0.14	57.31	0.20	37.58	0.15	57.66	0.21	7.51	1.32
AMDF (ours)	41.40	0.16	32.14	0.14	32.80	0.13	32.17	0.14	59.10	0.21	34.48	0.15	60.31	0.21	2.31	2.04

Accuracy vs. trainable parameters

Averaged over the five dense benchmarks, AMDF has the best mean rank with about a quarter of ConcatLayers' parameters. ConcatLayers still wins depth and camera pose.

	Mean rank $\downarrow$ (lower is better)	Trainable params (M)
AMDF (ours)	1.48	1.79
ConcatLayers	1.72	6.99
WeightedLayers	3.40	0.70
LastLayer	4.09	0.70
AvgLayer	4.31	0.70

Fig. 6: Mean of the per-table ranks over NYUv2 normals, depth, segmentation, Taskonomy and NAVI pose; parameters (probe + task head, ViT-L backbones) averaged over the same tasks. CUB-200 is excluded (different set of methods).